

2014 January C34 Q9

9. (a) Use the substitution $u = 4 - \sqrt{x}$ to find

$$\int \frac{dx}{4 - \sqrt{x}} \quad (6)$$

A team of scientists is studying a species of slow growing tree.

The rate of change in height of a tree in this species is modelled by the differential equation

$$\frac{dh}{dt} = \frac{4 - \sqrt{h}}{20}$$

where h is the height in metres and t is the time measured in years after the tree is planted.

- (b) Find the range in values of h for which the height of a tree in this species is increasing. (2)

- (c) Given that one of these trees is 1 metre high when it is planted, calculate the time it would take to reach a height of 10 metres. Write your answer to 3 significant figures. (7)

2014 January C34 Q12

12.

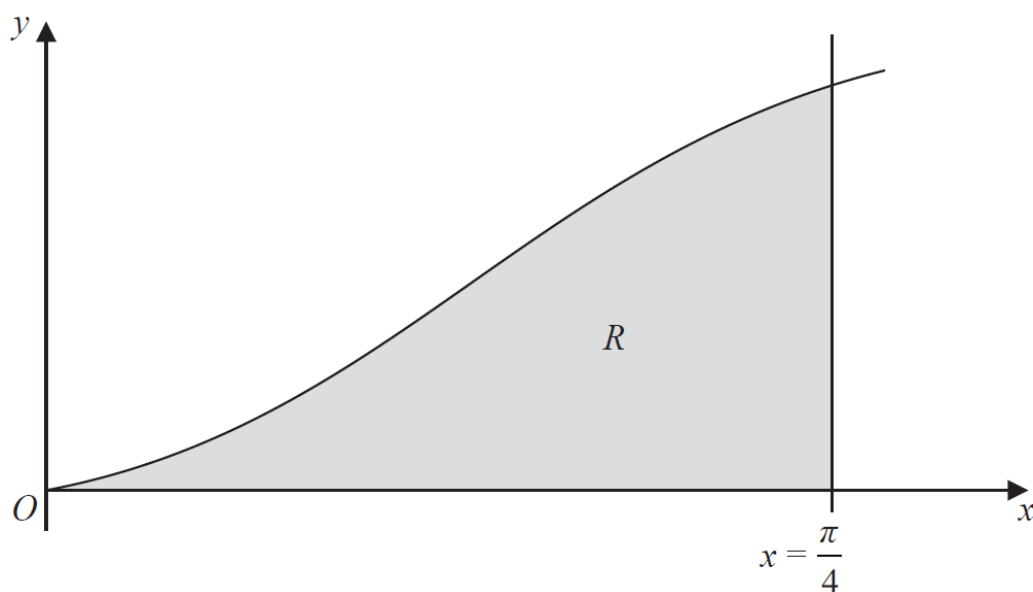


Figure 4

Figure 4 shows a sketch of part of the curve with equation

$$y = x(\sin x + \cos x), \quad 0 \leq x \leq \frac{\pi}{4}$$

The finite region R , shown shaded in Figure 4, is bounded by the curve, the x -axis and the line $x = \frac{\pi}{4}$. This shaded region is rotated through 2π radians about the x -axis to form a solid of revolution, with volume V .

(a) Assuming the formula for volume of revolution show that $V = \int_0^{\frac{\pi}{4}} \pi x^2 (1 + \sin 2x) dx$ (3)

(b) Hence using calculus find the exact value of V .

You must show your working.

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(9)

2014 June C34 Q6

6. (a) Express $\frac{5 - 4x}{(2x - 1)(x + 1)}$ in partial fractions.

(3)

- (b) (i) Find a general solution of the differential equation

$$(2x - 1)(x + 1)\frac{dy}{dx} = (5 - 4x)y, \quad x > \frac{1}{2}$$

Given that $y = 4$ when $x = 2$,

- (ii) find the particular solution of this differential equation.
Give your answer in the form $y = f(x)$.

(7)

2014 June C34 Q9

9.

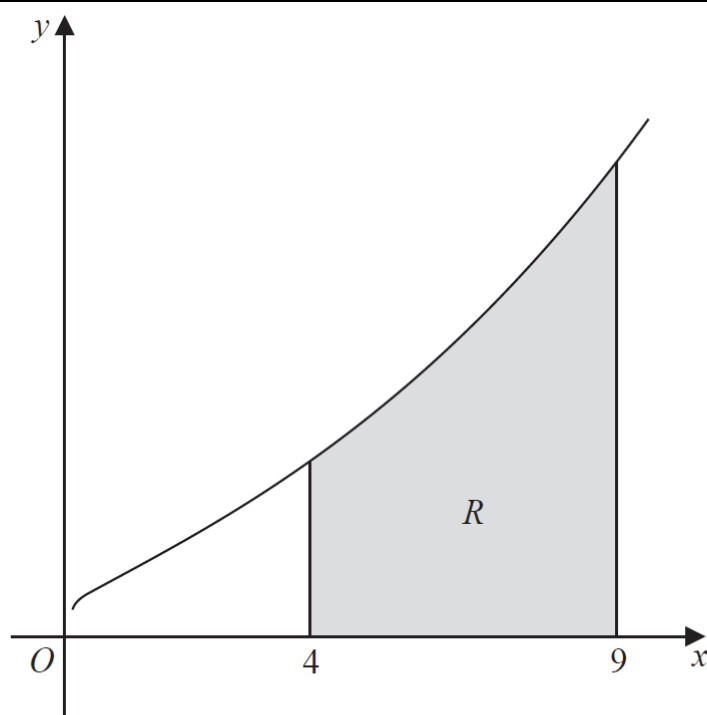


Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = e^{\sqrt{x}}$, $x > 0$

The finite region R , shown shaded in Figure 1, is bounded by the curve, the x -axis and the lines $x = 4$ and $x = 9$

- (a) Use the trapezium rule, with 5 strips of equal width, to obtain an estimate for the area of R , giving your answer to 2 decimal places. (4)
- (b) Use the substitution $u = \sqrt{x}$ to find, by integrating, the exact value for the area of R . (7)

12.

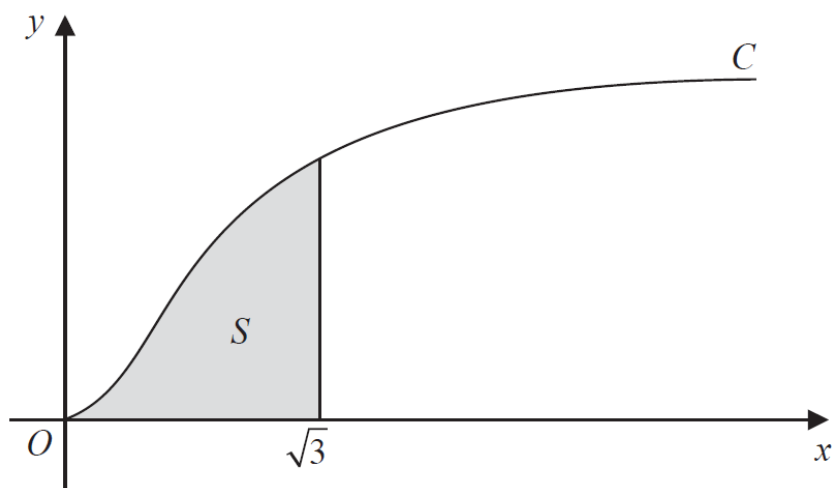


Figure 3

Figure 3 shows a sketch of part of the curve C with parametric equations

$$x = \tan t, \quad y = 2 \sin^2 t, \quad 0 \leq t < \frac{\pi}{2}$$

The finite region S , shown shaded in Figure 3, is bounded by the curve C , the line $x = \sqrt{3}$ and the x -axis. This shaded region is rotated through 2π radians about the x -axis to form a solid of revolution.

(a) Show that the volume of the solid of revolution formed is given by

$$4\pi \int_0^{\frac{\pi}{3}} (\tan^2 t - \sin^2 t) dt \quad (6)$$

(b) Hence use integration to find the exact value for this volume.

(6)

2015 January C34 Q4

4. Use the substitution $x = 2 \sin \theta$ to find the exact value of

$$\int_0^{\sqrt{3}} \frac{1}{(4-x^2)^{\frac{3}{2}}} dx$$

(7)

9.

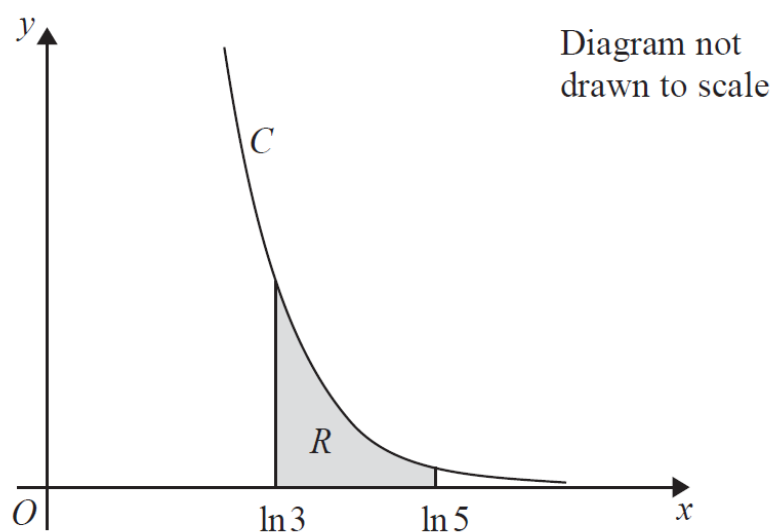


Figure 2

The curve C has parametric equations

$$x = \ln(t + 2), \quad y = \frac{4}{t^2} \quad t > 0$$

The finite region R , shown shaded in Figure 2, is bounded by the curve C , the x -axis and the lines with equations $x = \ln 3$ and $x = \ln 5$

(a) Show that the area of R is given by the integral

$$\int_1^3 \frac{4}{t^2(t+2)} dt \quad (3)$$

(b) Hence find an exact value for the area of R .

Write your answer in the form $(a + \ln b)$, where a and b are rational numbers.

(7)

2015 January C34 Q12

12.

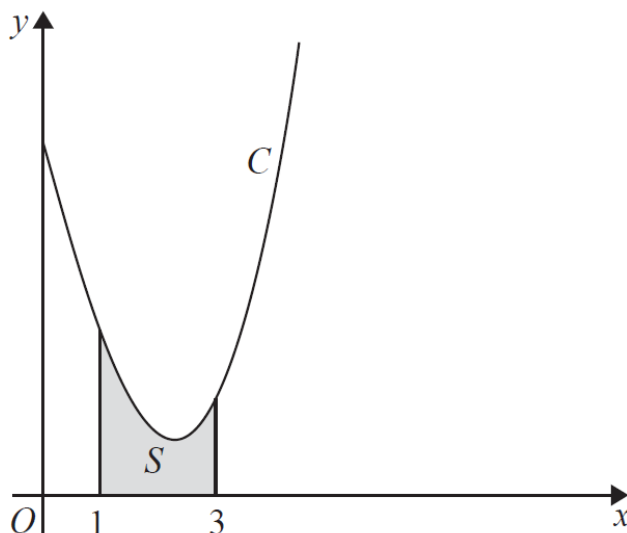


Figure 4

Figure 4 shows a sketch of part of the curve C with equation

$$y = \frac{x^2 \ln x}{3} - 2x + 4, \quad x > 0$$

The finite region S , shown shaded in Figure 4, is bounded by the curve C , the x -axis and the lines with equations $x = 1$ and $x = 3$

- (a) Complete the table below with the value of y corresponding to $x = 2$. Give your answer to 4 decimal places.

x	1	1.5	2	2.5	3
y	2	1.3041		0.9089	1.2958

(1)

- (b) Use the trapezium rule, with all the values of y in the completed table, to obtain an estimate for the area of S , giving your answer to 3 decimal places.

(3)

- (c) Use calculus to find the exact area of S .

Give your answer in the form $\frac{a}{b} + \ln c$, where a , b and c are integers.

(6)

2015 June C34 Q8

8. (a) Prove by differentiation that

$$\frac{d}{dy}(\ln \tan 2y) = \frac{4}{\sin 4y}, \quad 0 < y < \frac{\pi}{4}$$

(4)

- (b) Given that $y = \frac{\pi}{6}$ when $x = 0$, solve the differential equation

$$\frac{dy}{dx} = 2 \cos x \sin 4y, \quad 0 < y < \frac{\pi}{4}$$

Give your answer in the form $\tan 2y = Ae^{B \sin x}$, where A and B are constants to be determined.

(6)

13.

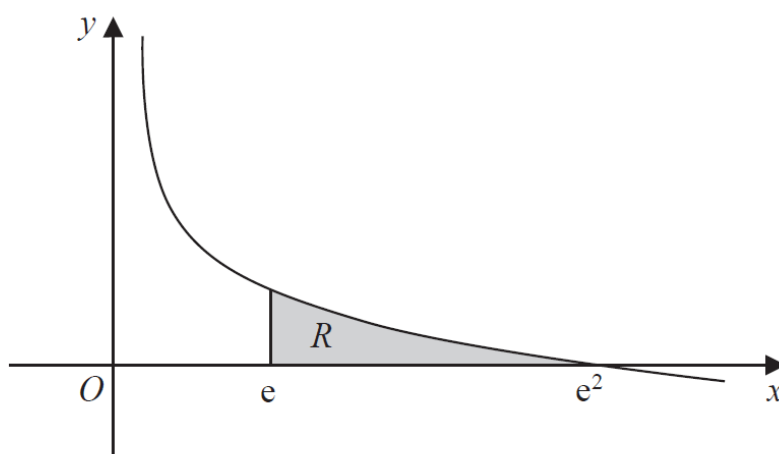


Figure 5

Figure 5 shows a sketch of part of the curve with equation $y = 2 - \ln x$, $x > 0$

The finite region R , shown shaded in Figure 5, is bounded by the curve, the x -axis and the line with equation $x = e$.

The table below shows corresponding values of x and y for $y = 2 - \ln x$

x	e	$\frac{e + e^2}{2}$	e^2
y	1		0

- (a) Complete the table giving the value of y to 4 decimal places. (1)
- (b) Use the trapezium rule, with all the values of y in the completed table, to obtain an estimate for the area of R , giving your answer to 3 decimal places. (3)
- (c) Use integration by parts to show that $\int (\ln x)^2 dx = x (\ln x)^2 - 2x \ln x + 2x + c$ (4)

The area R is rotated through 360° about the x -axis.

- (d) Use calculus to find the exact volume of the solid generated.

Write your answer in the form $\pi e(pe + q)$, where p and q are integers to be found.

(6)

4.

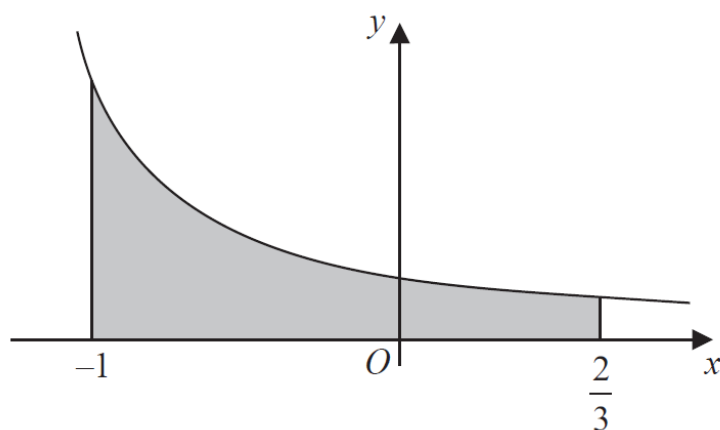


Figure 1

The curve C with equation $y = \frac{2}{(4 + 3x)}$, $x > -\frac{4}{3}$ is shown in Figure 1

The region bounded by the curve, the x -axis and the lines $x = -1$ and $x = \frac{2}{3}$, is shown shaded in Figure 1

This region is rotated through 360 degrees about the x -axis.

(a) Use calculus to find the exact value of the volume of the solid generated.

(5)

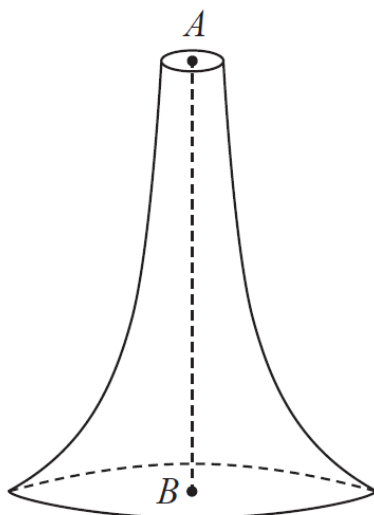


Figure 2

Figure 2 shows a candle with axis of symmetry AB where $AB = 15$ cm. A is a point at the centre of the top surface of the candle and B is a point at the centre of the base of the candle. The candle is geometrically similar to the solid generated in part (a).

(b) Find the volume of this candle.

(2)

2016 January C34 Q8

8.

$$f(\theta) = 9\cos^2\theta + \sin^2\theta$$

(a) Show that $f(\theta) = a + b\cos 2\theta$, where a and b are integers which should be found.

(3)

(b) Using your answer to part (a) and integration by parts, find the exact value of

$$\int_0^{\frac{\pi}{2}} \theta^2 f(\theta) d\theta$$

(6)

2016 January C34 Q9

9. (a) Express $\frac{3x^2 - 4}{x^2(3x - 2)}$ in partial fractions.

(4)

- (b) Given that $x > \frac{2}{3}$, find the general solution of the differential equation

$$x^2(3x - 2) \frac{dy}{dx} = y(3x^2 - 4)$$

Give your answer in the form $y = f(x)$.

(6)

2016 June C34 Q5

5. Use integration by parts to find the exact value of

$$\int_0^2 x2^x dx$$

Write your answer as a single simplified fraction.

(6)

12.

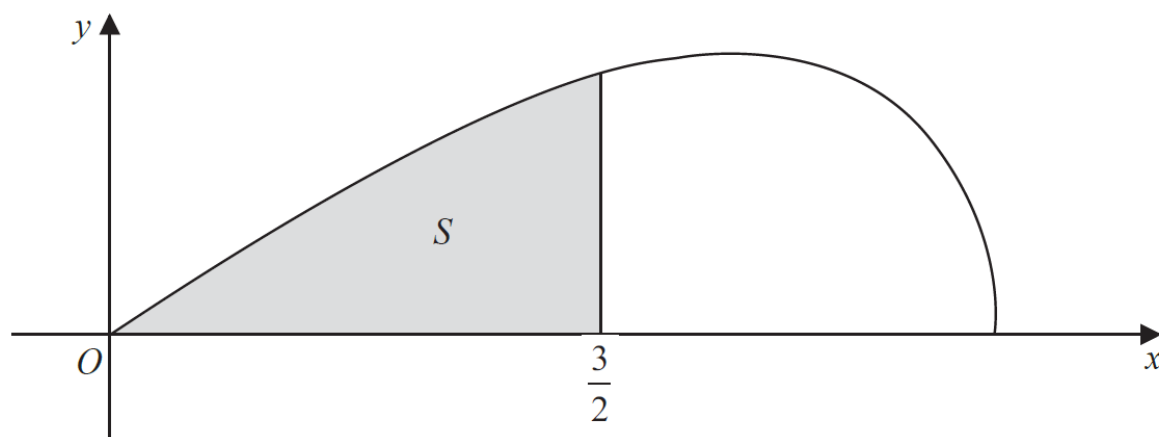


Figure 3

Figure 3 shows a sketch of the curve with parametric equations

$$x = 3 \sin t, \quad y = 2 \sin 2t, \quad 0 \leq t \leq \frac{\pi}{2}$$

The finite region S , shown shaded in Figure 3, is bounded by the curve, the x -axis and the line with equation $x = \frac{3}{2}$

The shaded region S is rotated through 2π radians about the x -axis to form a solid of revolution.

(a) Show that the volume of the solid of revolution is given by

$$k \int_0^a \sin^2 t \cos^3 t \, dt$$

where k and a are constants to be given in terms of π .

(5)

(b) Use the substitution $u = \sin t$, or otherwise, to find the exact value of this volume, giving your answer in the form $\frac{p\pi}{q}$ where p and q are integers.

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(6)

5.

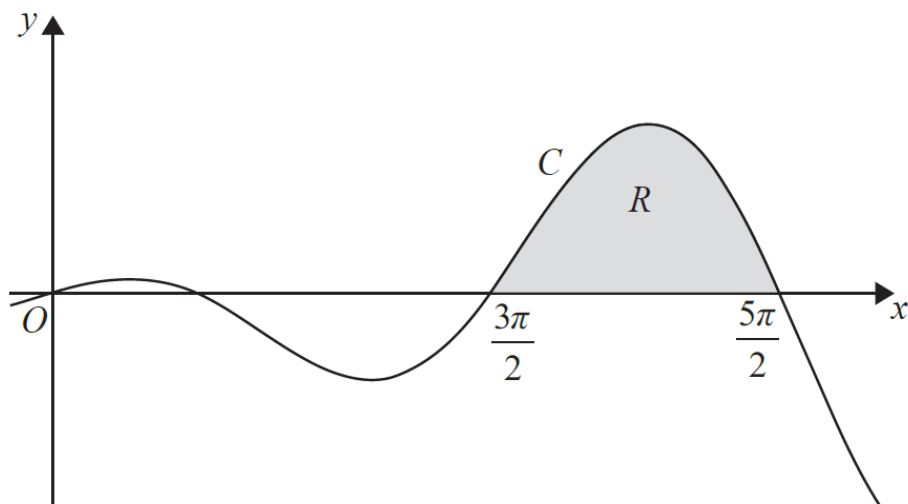


Figure 1

Figure 1 shows a sketch of part of the curve C with equation

$$y = x \cos x, \quad x \in \mathbb{R}$$

The finite region R , shown shaded in Figure 1, is bounded by the curve C and the x -axis for $\frac{3\pi}{2} \leq x \leq \frac{5\pi}{2}$

- (a) Complete the table below with the exact value of y corresponding to $x = \frac{7\pi}{4}$ and with the exact value of y corresponding to $x = \frac{9\pi}{4}$

x	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π	$\frac{9\pi}{4}$	$\frac{5\pi}{2}$
y	0		2π		0

(1)

- (b) Use the trapezium rule, with all five y values in the completed table, to find an approximate value for the area of R , giving your answer to 4 significant figures.

(3)

- (c) Find

$$\int x \cos x \, dx$$

(3)

- (d) Using your answer from part (c), find the exact area of the region R .

(2)

2017 January C34 Q9

9.

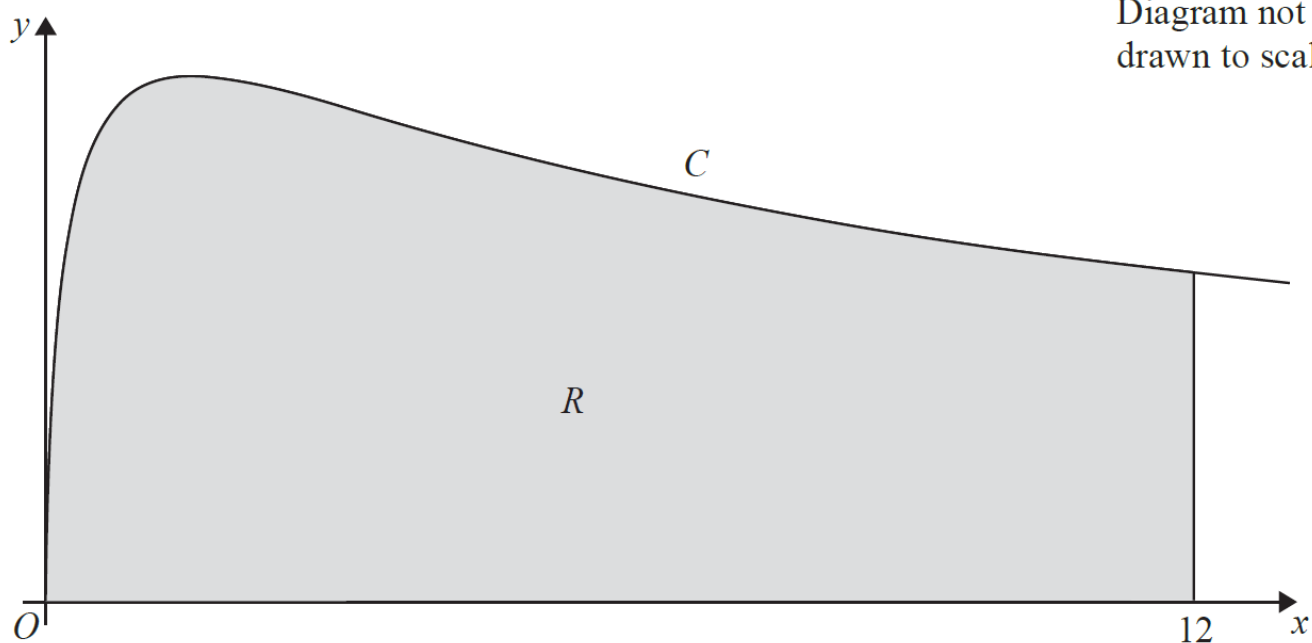


Figure 3

(a) By using the substitution $u = 2x + 3$, show that

$$\int_0^{12} \frac{x}{(2x+3)^2} dx = \frac{1}{2} \ln 3 - \frac{2}{9} \quad (7)$$

The curve C has equation

$$y = \frac{9\sqrt{x}}{(2x+3)}, \quad x > 0$$

The finite region R , shown shaded in Figure 3, is bounded by the curve C , the x -axis and the line with equation $x = 12$. The region R is rotated through 2π radians about the x -axis to form a solid of revolution.

(b) Use the result of part (a) to find the exact value of the volume of the solid generated. (2)

2017 January C34 Q12

12. In freezing temperatures, ice forms on the surface of the water in a barrel. At time t hours after the start of freezing, the thickness of the ice formed is x mm. You may assume that the thickness of the ice is uniform across the surface of the water.

At 4pm there is no ice on the surface, and freezing begins.

At 6pm, after two hours of freezing, the ice is 1.5 mm thick.

In a simple model, the rate of increase of x , in mm per hour, is assumed to be constant for a period of 20 hours.

Using this simple model,

(a) express t in terms of x , (2)

(b) find the value of t when $x = 3$ (1)

In a second model, the rate of increase of x , in mm per hour, is given by the differential equation

$$\frac{dx}{dt} = \frac{\lambda}{(2x + 1)} \text{ where } \lambda \text{ is a constant and } 0 \leq t \leq 20$$

Using this second model,

(c) solve the differential equation and express t in terms of x and λ , (3)

(d) find the exact value for λ , (1)

(e) find at what time the ice is predicted to be 3 mm thick. (2)

2017 June C34 Q2

2. Use integration by parts to find the exact value of $\int_1^e \frac{\ln x}{x^2} dx$

Write your answer in the form $a + \frac{b}{e}$, where a and b are integers.

(6)

2017 June C34 Q8

8.

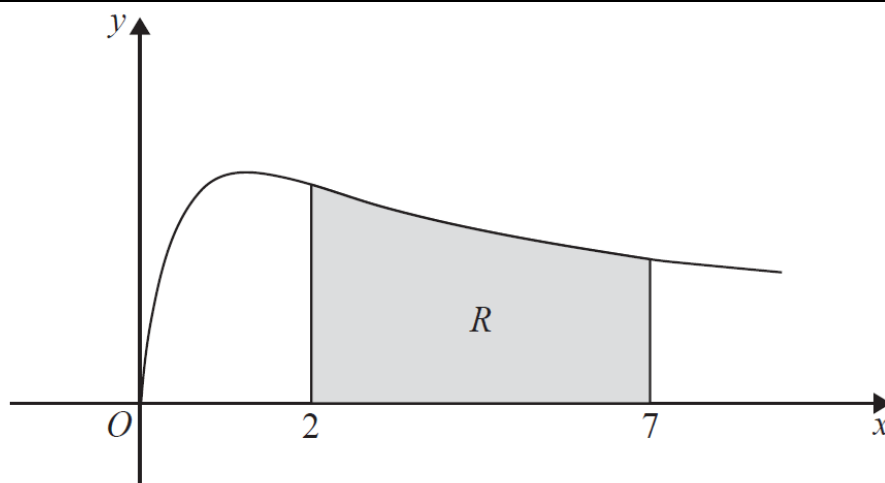


Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = \sqrt{\frac{x}{x^2 + 1}}$, $x \geq 0$

The finite region R , shown shaded in Figure 1, is bounded by the curve, the line with equation $x = 2$, the x -axis and the line with equation $x = 7$

The table below shows corresponding values of x and y for $y = \sqrt{\frac{x}{x^2 + 1}}$

x	2	3	4	5	6	7
y	0.6325	0.5477	0.4851	0.4385	0.4027	0.3742

- (a) Use the trapezium rule, with all the values of y in the table, to find an estimate for the area of R , giving your answer to 3 decimal places. (3)

The region R is rotated 360° about the x -axis to form a solid of revolution.

- (b) Use calculus to find the exact volume of the solid of revolution formed. Write your answer in its simplest form. (4)

14.

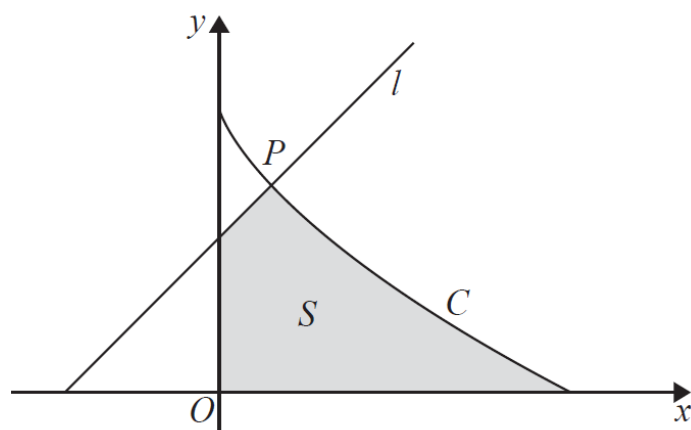


Figure 6

Figure 6 shows a sketch of the curve C with parametric equations

$$x = 8 \cos^3 \theta, \quad y = 6 \sin^2 \theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

Given that the point P lies on C and has parameter $\theta = \frac{\pi}{3}$

(a) find the coordinates of P .

(2)

The line l is the normal to C at P .

(b) Show that an equation of l is $y = x + 3.5$

(5)

The finite region S , shown shaded in Figure 6, is bounded by the curve C , the line l , the y -axis and the x -axis.

(c) Show that the area of S is given by

$$4 + 144 \int_0^{\frac{\pi}{3}} (\sin \theta \cos^2 \theta - \sin \theta \cos^4 \theta) \, d\theta$$

(6)

(d) Hence, by integration, find the exact area of S .

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(3)

2017 October C34 Q8

8. Use partial fractions, and integration, to find the exact value of $\int_3^4 \frac{2x^2 - 3}{x(x-1)} dx$

Write your answer in the form $a + \ln b$, where a is an integer and b is a rational constant.

(8)

2017 October C34 Q11

11. (a) Given $0 \leq h < 25$, use the substitution $u = 5 - \sqrt{h}$ to show that

$$\int \frac{dh}{5 - \sqrt{h}} = -10 \ln(5 - \sqrt{h}) - 2\sqrt{h} + k$$

where k is a constant.

(6)

A team of scientists is studying a species of tree.

The rate of change in height of a tree of this species is modelled by the differential equation

$$\frac{dh}{dt} = \frac{t^{0.2}(5 - \sqrt{h})}{5}$$

where h is the height of the tree in metres and t is the time in years after the tree is planted.

One of these trees is 2 metres high when it is planted.

- (b) Use integration to calculate the time it would take for this tree to reach a height of 15 metres, giving your answer to one decimal place.

(7)

- (c) Hence calculate the rate of change in height of this tree when its height is 15 metres. Write your answer in centimetres per year to the nearest centimetre.

(1)

2018 January C34 Q5

5. (a) Express $\frac{9(4+x)}{16-9x^2}$ in partial fractions. (3)

Given that

$$f(x) = \frac{9(4+x)}{16-9x^2}, \quad x \in \mathbb{R}, \quad -\frac{4}{3} < x < \frac{4}{3}$$

- (b) express $\int f(x)dx$ in the form $\ln(g(x))$, where $g(x)$ is a rational function. (4)

2018 January C34 Q6

6.

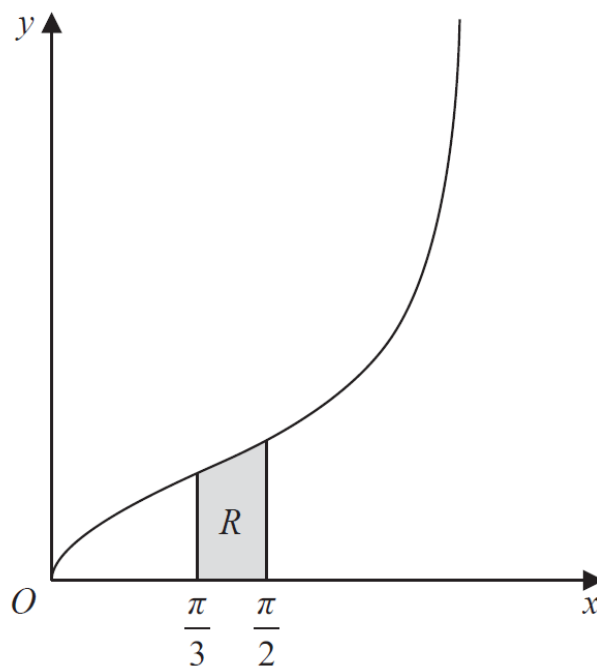


Figure 2

The curve shown in Figure 2 has equation

$$y^2 = 3 \tan\left(\frac{x}{2}\right), \quad 0 < x < \pi, \quad y > 0$$

The finite region R , shown shaded in Figure 2, is bounded by the curve, the line with equation $x = \frac{\pi}{3}$ the x -axis and the line with equation $x = \frac{\pi}{2}$

The region R is rotated through 360° about the x -axis to generate a solid of revolution.

Show that the exact value of the volume of the solid generated may be written as $A \ln\left(\frac{3}{2}\right)$, where A is a constant to be found.

(5)

2018 January C34 Q13

13.

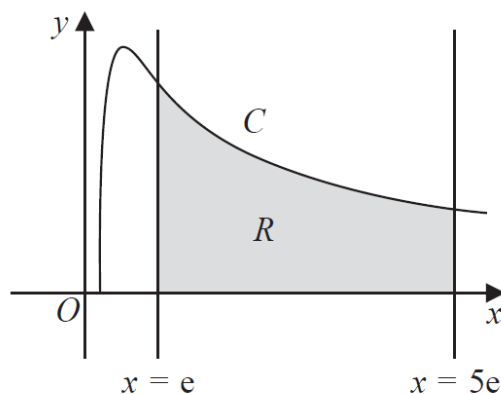


Figure 4

Figure 4 shows a sketch of part of the curve C with equation

$$y = \frac{1}{2x} \ln 2x, \quad x > \frac{1}{2}$$

The finite region R , shown shaded in Figure 4, is bounded by the curve C , the x -axis and the lines with equations $x = e$ and $x = 5e$.

The table below shows corresponding values of x and y for $y = \frac{1}{2x} \ln 2x$. The values for y are given to 4 significant figures.

x	e	$2e$	$3e$	$4e$	$5e$
y	0.3114	0.2195	0.1712	0.1416	0.1215

- (a) Use the trapezium rule with all the y values in the table to find an approximate value for the area of R , giving your answer to 3 significant figures.

(3)

- (b) Using the substitution $u = \ln 2x$, or otherwise, find $\int \frac{1}{2x} \ln 2x \, dx$

(3)

- (c) Use your answer to part (b) to find the true area of R , giving your answer to 3 significant figures.

(2)

- (d) Using calculus, find an equation for the tangent to the curve at the point where $x = \frac{e^2}{2}$, giving your answer in the form $y = mx + c$ where m and c are exact multiples of powers of e .

(5)

2018 January C34 Q14

14. The volume of a spherical balloon of radius r cm is V cm³, where $V = \frac{4}{3}\pi r^3$

(a) Find $\frac{dV}{dr}$

(1)

The volume of the balloon increases with time t seconds according to the formula

$$\frac{dV}{dt} = \frac{9000\pi}{(t+81)^{\frac{5}{4}}} \quad t \geq 0$$

(b) Using the chain rule, or otherwise, show that

$$\frac{dr}{dt} = \frac{k}{r^n(t+81)^{\frac{5}{4}}} \quad t \geq 0$$

where k and n are constants to be found.

(2)

Initially, the radius of the balloon is 3 cm.

(c) Using the values of k and n found in part (b), solve the differential equation

$$\frac{dr}{dt} = \frac{k}{r^n(t+81)^{\frac{5}{4}}} \quad t \geq 0$$

to obtain a formula for r in terms of t .

(6)

(d) Hence find the radius of the balloon when $t = 175$, giving your answer to 3 significant figures.

(1)

(e) Find the rate of increase of the radius of the balloon when $t = 175$. Give your answer to 3 significant figures.

(2)

2018 June C34 Q1

1. (i) Find

$$\int \frac{2x^2 + 5x + 1}{x^2} dx, \quad x > 0$$

(3)

(ii) Find

$$\int x \cos 2x dx$$

(3)

2018 June C34 Q9

9.

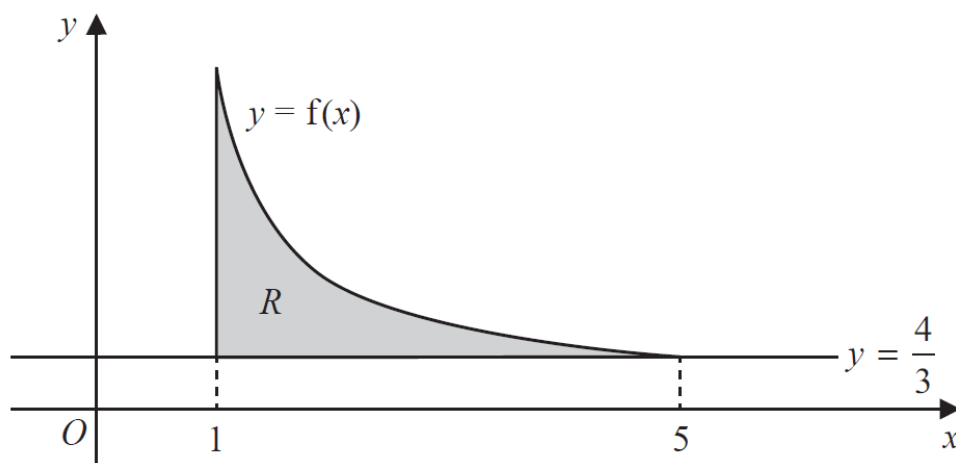


Diagram not
drawn to scale

Figure 2

(a) Find

$$\int \frac{1}{(2x-1)^2} dx$$

(2)

Figure 2 shows a sketch of the curve with equation $y = f(x)$ where

$$f(x) = \frac{12}{(2x-1)} \quad 1 \leq x \leq 5$$

The finite region R , shown shaded in Figure 2, is bounded by the line with equation $x = 1$, the curve with equation $y = f(x)$ and the line with equation $y = \frac{4}{3}$.

The region R is rotated through 2π radians about the x -axis to form a solid of revolution.

(b) Find the exact value of the volume of the solid generated, giving your answer in its simplest form.

(6)

11.

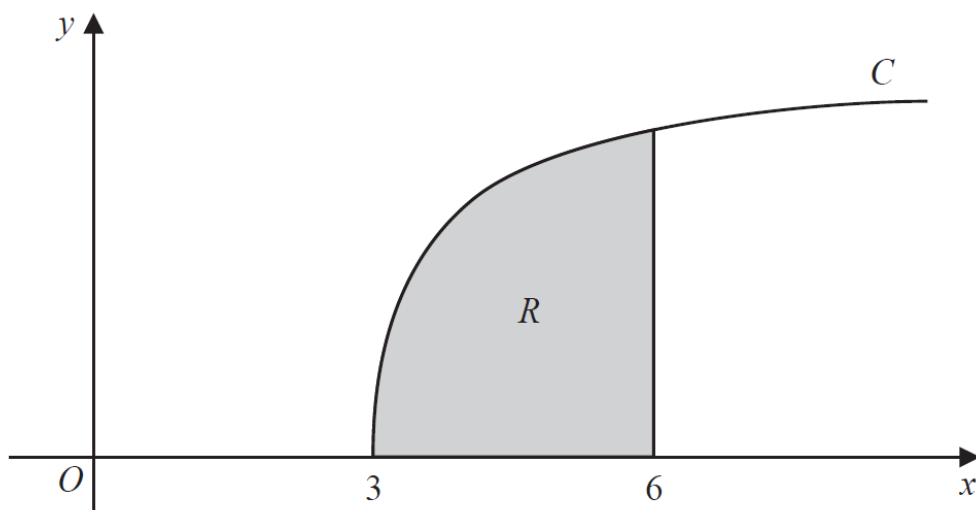


Figure 3

- (a) By writing $\sec \theta$ as $\frac{1}{\cos \theta}$, show that when $x = 3 \sec \theta$,

$$\frac{dx}{d\theta} = 3 \sec \theta \tan \theta$$

(2)

Figure 3 shows a sketch of part of the curve C with equation

$$y = \frac{\sqrt{x^2 - 9}}{x} \quad x \geq 3$$

The finite region R , shown shaded in Figure 3, is bounded by the curve C , the x -axis and the line with equation $x = 6$

- (b) Use the substitution $x = 3 \sec \theta$ to find the exact value of the area of R .

[Solutions based entirely on graphical or numerical methods are not acceptable.]

(7)

2018 June C34 Q13

13. (a) Express $\frac{1}{(4-x)(2-x)}$ in partial fractions. (2)

The mass, x grams, of a substance at time t seconds after a chemical reaction starts is modelled by the differential equation

$$\frac{dx}{dt} = k(4-x)(2-x), \quad t \geq 0, \quad 0 \leq x < 2$$

where k is a constant.

Given that when $t = 0$, $x = 0$

- (b) solve the differential equation and show that the solution can be written as

$$x = \frac{4 - 4e^{2kt}}{1 - 2e^{2kt}}$$

(7)

Given that $k = 0.1$

- (c) find the value of t when $x = 1$, giving your answer, in seconds, to 3 significant figures. (2)

2018 October C34 Q5

5.

$$f(x) = \frac{4x^2 + 5x + 3}{(x+2)(1-x)^2} \equiv \frac{A}{(x+2)} + \frac{B}{(1-x)} + \frac{C}{(1-x)^2}$$

(a) Find the values of the constants A , B and C .

(4)

(b) (i) Hence find $\int f(x) \, dx$.

(ii) Find the exact value of $\int_0^{\frac{1}{2}} f(x) \, dx$, writing your answer in the form $p + \ln q$, where p and q are constants.

(6)

2018 October C34 Q8

8. (i) Find $\int x \sin x \, dx$

(3)

(ii) (a) Use the substitution $x = \sec \theta$ to show that

$$\int_1^2 \sqrt{1 - \frac{1}{x^2}} \, dx = \int_0^{\frac{\pi}{3}} \tan^2 \theta \, d\theta$$

(3)

(b) Hence find the exact value of

$$\int_1^2 \sqrt{1 - \frac{1}{x^2}} \, dx$$

(4)

2019 January C34 Q7

7.

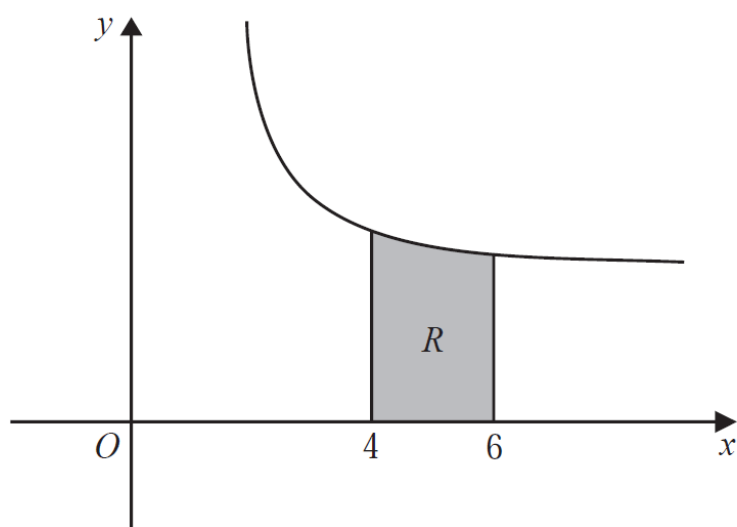


Figure 1

Figure 1 shows a sketch of part of the curve with equation

$$y = \frac{x+7}{\sqrt{2x-3}} \quad x > \frac{3}{2}$$

The region R , shown shaded in Figure 1, is bounded by the curve, the line with equation $x = 4$, the x -axis and the line with equation $x = 6$

- (a) Use the trapezium rule with 4 strips of equal width to find an estimate for the area of R , giving your answer to 2 decimal places. (4)
- (b) Using the substitution $u = 2x - 3$, or otherwise, use calculus to find the exact area of R , giving your answer in the form $a + b\sqrt{5}$, where a and b are constants to be found. (7)

2019 January C34 Q9

9.

(a) Find $\int x \sin 2x \, dx$

(3)

(b) Find $\int (x + \sin 2x)^2 \, dx$

(4)

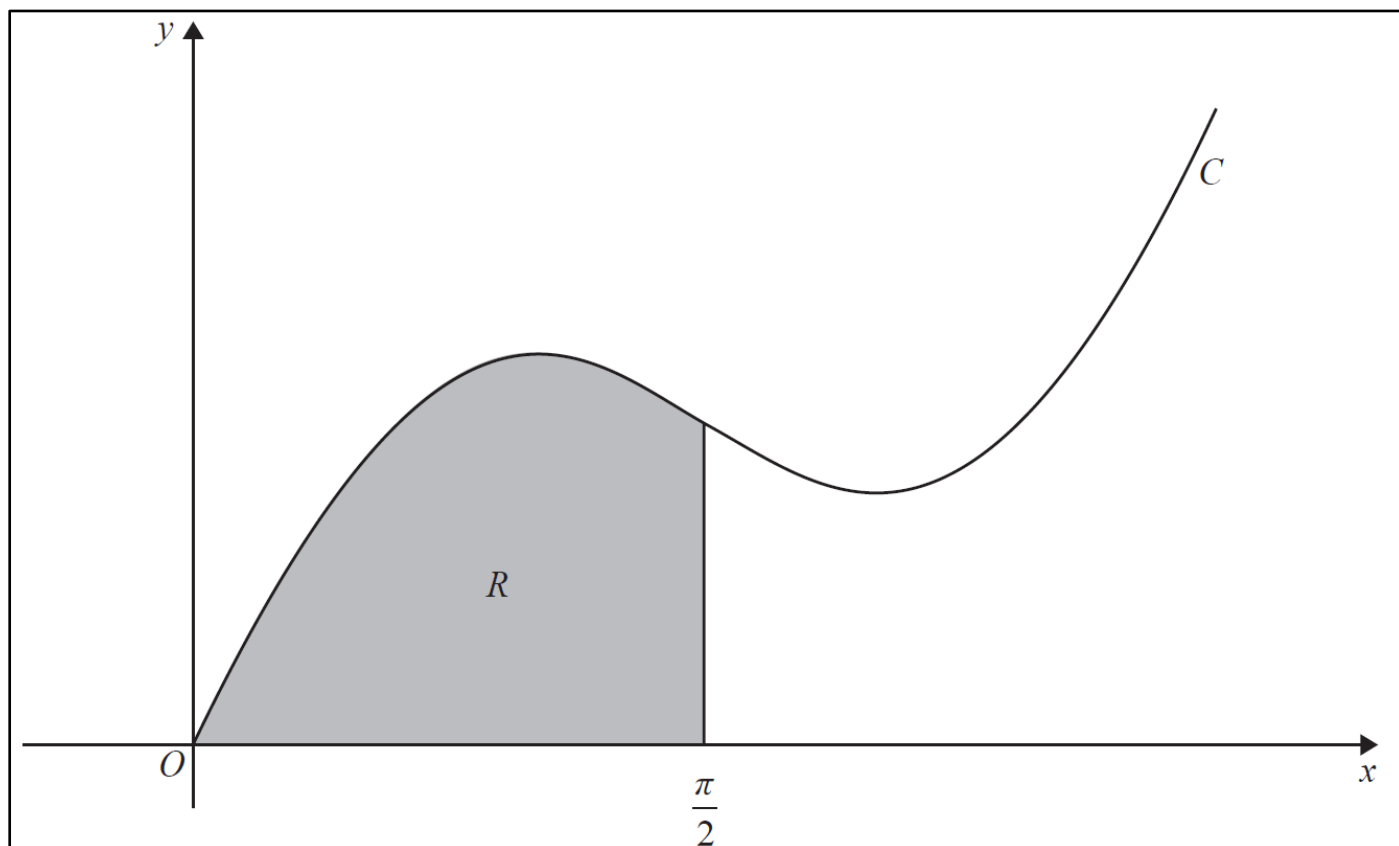


Figure 2

Figure 2 shows a sketch of part of the curve C with equation $y = x + \sin 2x$.

The region R , shown shaded in Figure 2, is bounded by C , the x -axis and the line with equation $x = \frac{\pi}{2}$

The region R is rotated through 2π radians about the x -axis to form a solid of revolution.

(c) Find the exact value for the volume of this solid, giving your answer as a single, simplified fraction.

(3)

2019 June C34 Q2

2. (a) Find $\int \frac{4x+3}{x} dx$, $x > 0$ (2)

(b) Given that $y = 25$ at $x = 1$, solve the differential equation

$$\frac{dy}{dx} = \frac{(4x+3)y^{\frac{1}{2}}}{x} \quad x > 0, y > 0$$

giving your answer in the form $y = [g(x)]^2$.

(5)

2019 June C34 Q9

9. (a) Using the formula for $\sin(A + B)$ and the relevant double angle formulae, find an identity for $\sin 3x$, giving your answer in the form

$$\sin(3x) \equiv P \sin x + Q \sin^3 x$$

where P and Q are constants to be determined.

(4)

- (b) Hence, showing each step of your working, evaluate

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \sin 3x \cos x \, dx$$

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(4)

2019 June C34 Q11

11. (a) Given

$$\frac{9}{t^2(t-3)} \equiv \frac{A}{t} + \frac{B}{t^2} + \frac{C}{(t-3)}$$

find the value of the constants A , B and C .

(3)

(b)

$$I = \int_4^{12} \frac{9}{t^2(t-3)} dt$$

Find the exact value of I , giving your answer in the form $\ln(a) - b$, where a and b are positive constants.

(6)

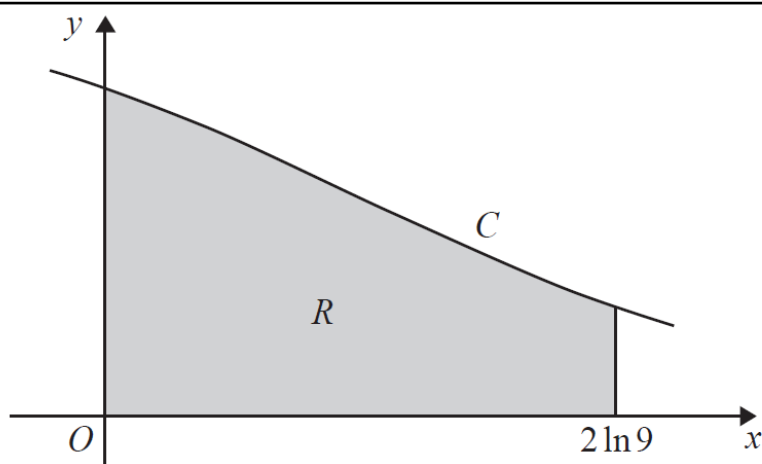


Figure 3

Figure 3 shows a sketch of part of the curve C with parametric equations

$$x = 2 \ln(t-3), \quad y = \frac{6}{t} \quad t > 3$$

The finite region R , shown shaded in Figure 3, is bounded by the curve C , the y -axis, the x -axis and the line with equation $x = 2 \ln 9$

The region R is rotated 360° about the x -axis to form a solid of revolution.

(c) Show that the exact volume of the solid generated is

$$k \times I$$

where k is a constant to be found.

(3)

2019 June C34 Q13

13.

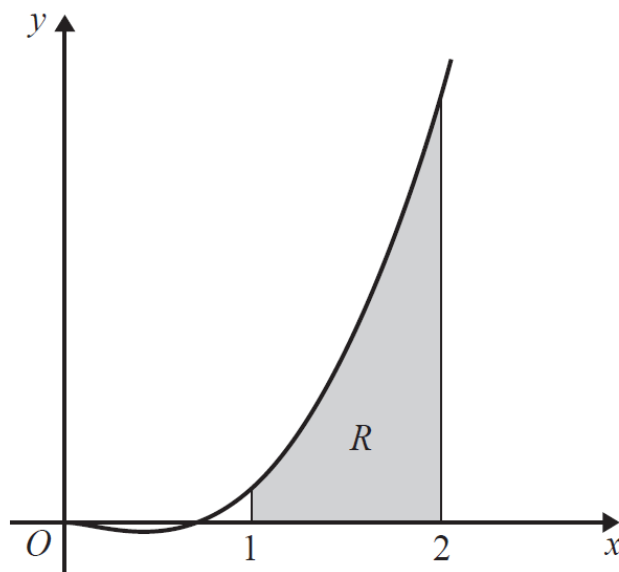


Figure 4

Figure 4 shows a sketch of the curve with equation $y = 12x^2 \ln(2x^2)$, $x > 0$

The finite region R , shown shaded in Figure 4, is bounded by the curve, the line with equation $x = 1$, the x -axis and the line with equation $x = 2$

The table below shows corresponding values of x and y for $y = 12x^2 \ln(2x^2)$, with the values of y given to 3 significant figures.

x	1	1.25	1.5	1.75	2
y	8.32	21.4	40.6	66.6	99.8

- (a) Use the trapezium rule, with all the values of y , to obtain an estimate for the area of R , giving your answer to 2 significant figures.

(3)

- (b) Use the substitution $u = x^2$ to show that the area of R is given by

$$\int_1^4 6u^{\frac{1}{2}} \ln(2u) \, du$$

(3)

- (c) Hence, using calculus, find the exact area of R , writing your answer in the form $a + b \ln 2$, where a and b are constants to be found.

(6)

2020 October P4 Q3

3.

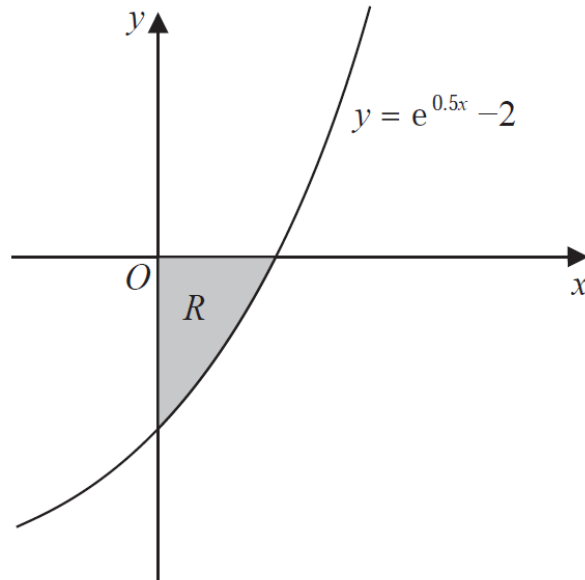


Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = e^{0.5x} - 2$

The region R , shown shaded in Figure 1, is bounded by the curve, the x -axis and the y -axis.

The region R is rotated 360° about the x -axis to form a solid of revolution.

Show that the volume of this solid can be written in the form $a \ln 2 + b$, where a and b are constants to be found.

(6)

5.

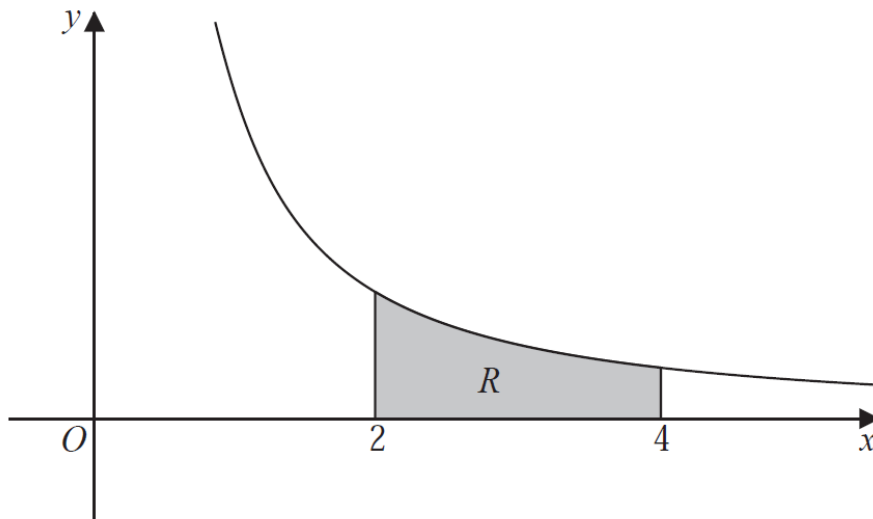


Figure 3

(a) Find $\int \frac{\ln x}{x^2} dx$

(3)

Figure 3 shows a sketch of part of the curve with equation

$$y = \frac{3 + 2x + \ln x}{x^2} \quad x > 0.5$$

The finite region R , shown shaded in Figure 3, is bounded by the curve, the line with equation $x = 2$, the x -axis and the line with equation $x = 4$

(b) Use the answer to part (a) to find the exact area of R , writing your answer in simplest form.

(4)

2020 October P4 Q7

7. (i) Using a suitable substitution, find, using calculus, the value of

$$\int_1^5 \frac{3x}{\sqrt{2x-1}} \, dx$$

(Solutions relying entirely on calculator technology are not acceptable.)

(6)

- (ii) Find

$$\int \frac{6x^2 - 16}{(x+1)(2x-3)} \, dx$$

(6)

2020 October P4 Q9

9. Bacteria are growing on the surface of a dish in a laboratory.

The area of the dish, $A \text{ cm}^2$, covered by the bacteria, t days after the bacteria start to grow, is modelled by the differential equation

$$\frac{dA}{dt} = \frac{A^{\frac{3}{2}}}{5t^2} \quad t > 0$$

Given that $A = 2.25$ when $t = 3$

- (a) show that

$$A = \left(\frac{pt}{qt + r} \right)^2$$

where p , q and r are integers to be found.

(7)

According to the model, there is a limit to the area that will be covered by the bacteria.

- (b) Find the value of this limit.

(2)

2021 January P4 Q5

5. **In this question you should show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

Using the substitution $u = 3 + \sqrt{2x - 1}$ find the exact value of

$$\int_1^{13} \frac{4}{3 + \sqrt{2x - 1}} dx$$

giving your answer in the form $p + q \ln 2$, where p and q are integers to be found.

(8)

7.

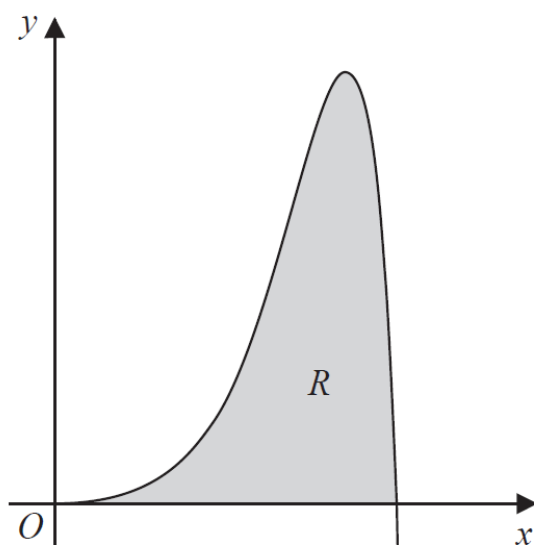


Figure 2

(a) Find $\int e^{2x} \sin x \, dx$

(5)

Figure 2 shows a sketch of part of the curve with equation

$$y = e^{2x} \sin x \quad x \geq 0$$

The finite region R is bounded by the curve and the x -axis and is shown shaded in Figure 2.

(b) Show that the exact area of R is $\frac{e^{2\pi} + 1}{5}$

(Solutions relying on calculator technology are not acceptable.)

(2)

2021 January P4 Q9

9.

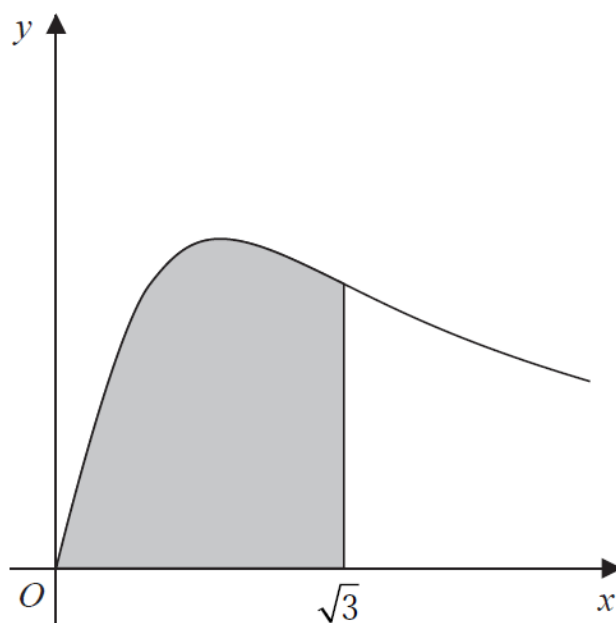


Figure 3

Figure 3 shows a sketch of part of the curve with parametric equations

$$x = \tan \theta \quad y = 2 \sin 2\theta \quad \theta \geq 0$$

The finite region, shown shaded in Figure 3, is bounded by the curve, the x -axis and the line with equation $x = \sqrt{3}$

The region is rotated through 2π radians about the x -axis to form a solid of revolution.

- (a) Show that the exact volume of this solid of revolution is given by

$$\int_0^k p(1 - \cos 2\theta) \, d\theta$$

where p and k are constants to be found.

(7)

- (b) Hence find, by algebraic integration, the exact volume of this solid of revolution.

(3)

2021 January P4 Q10

10. (a) Write $\frac{1}{(H-5)(H+3)}$ in partial fraction form.

(3)

The depth of water in a storage tank is being monitored.

The depth of water in the tank, H metres, is modelled by the differential equation

$$\frac{dH}{dt} = -\frac{(H-5)(H+3)}{40}$$

where t is the time, in days, from when monitoring began.

Given that the initial depth of water in the tank was 13 m,

- (b) solve the differential equation to show that

$$H = \frac{10 + 3e^{-0.2t}}{2 - e^{-0.2t}}$$

(7)

- (c) Hence find the time taken for the depth of water in the tank to fall to 8 m.

(Solutions relying entirely on calculator technology are not acceptable.)

(3)

According to the model, the depth of water in the tank will eventually fall to k metres.

- (d) State the value of the constant k .

(1)

2021 June P4 Q2

2.

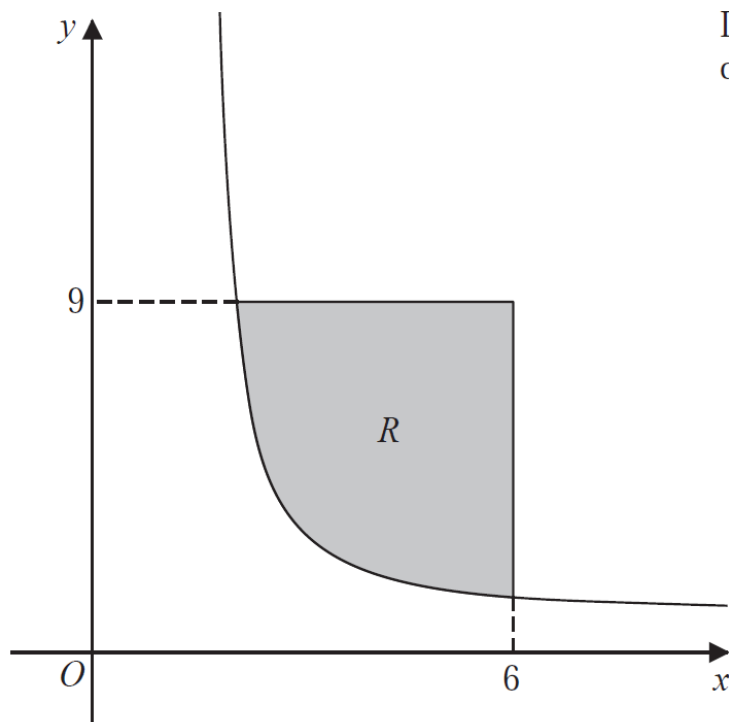


Figure 1

Figure 1 shows a sketch of part of the curve with equation

$$y = \frac{9}{(2x - 3)^{1.25}} \quad x > \frac{3}{2}$$

The finite region R , shown shaded in Figure 1, is bounded by the curve, the line with equation $y = 9$ and the line with equation $x = 6$

This region is rotated through 2π radians about the x -axis to form a solid of revolution.

Find, by algebraic integration, the exact volume of the solid generated.

(7)

2021 June P4 Q4

4. Use algebraic integration and the substitution $u = \sqrt{x}$ to find the exact value of

$$\int_1^4 \frac{10}{5x + 2x\sqrt{x}} dx$$

Write your answer in the form $4 \ln\left(\frac{a}{b}\right)$, where a and b are integers to be found.

(Solutions relying entirely on calculator technology are not acceptable.)

(8)

2021 June P4 Q8

8. (a) Given that $y = 1$ at $x = 0$, solve the differential equation

$$\frac{dy}{dx} = \frac{6xy^{\frac{1}{3}}}{e^{2x}} \quad y \geq 0$$

giving your answer in the form $y^2 = g(x)$.

(7)

- (b) Hence find the equation of the horizontal asymptote to the curve with equation $y^2 = g(x)$.

(2)

2021 October P4 Q2

2. Find the particular solution of the differential equation

$$\frac{dy}{dx} = \frac{4y^2}{\sqrt{4x+5}} \quad x > -\frac{5}{4}$$

for which $y = \frac{1}{3}$ at $x = -\frac{1}{4}$ giving your answer in the form $y = f(x)$

(6)

2021 October P4 Q6

6. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

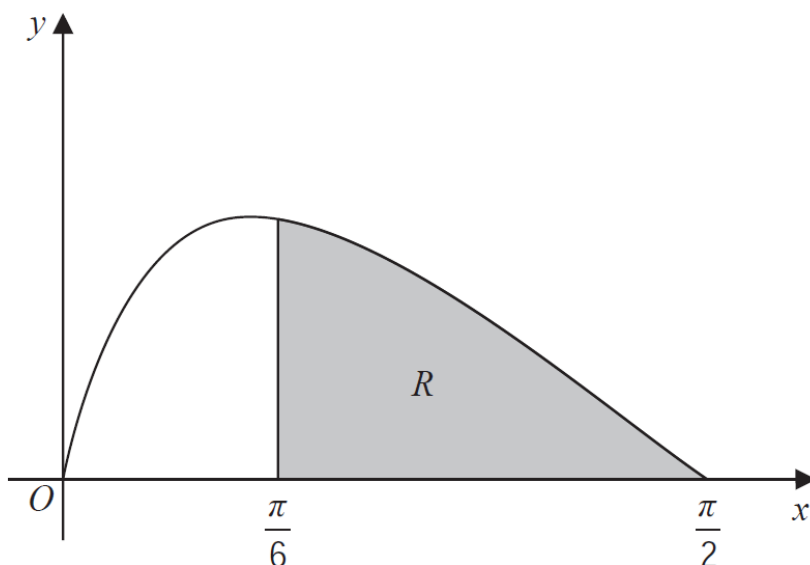


Figure 2

Figure 2 shows a sketch of the curve with equation

$$y = \frac{16 \sin 2x}{(3 + 4 \sin x)^2} \quad 0 \leq x \leq \frac{\pi}{2}$$

The region R , shown shaded in Figure 2, is bounded by the curve, the x -axis and the line with equation $x = \frac{\pi}{6}$

Using the substitution $u = 3 + 4 \sin x$, show that the area of R can be written in the form $a + \ln b$, where a and b are rational constants to be found.

(7)

2021 October P4 Q8

8. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

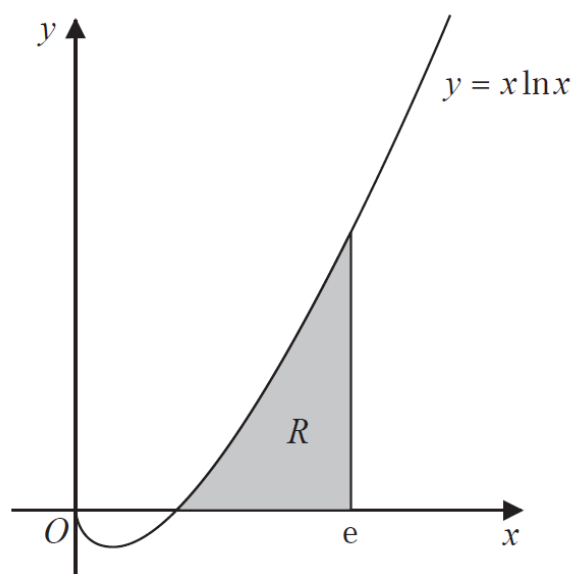


Figure 3

(a) Find $\int x^2 \ln x \, dx$

(3)

Figure 3 shows a sketch of part of the curve with equation

$$y = x \ln x \quad x > 0$$

The region R , shown shaded in Figure 3, lies entirely above the x -axis and is bounded by the curve, the x -axis and the line with equation $x = e$.

This region is rotated through 2π radians about the x -axis to form a solid of revolution.

(b) Find the exact volume of the solid formed, giving your answer in simplest form.

(4)

2022 January P4 Q5

5.

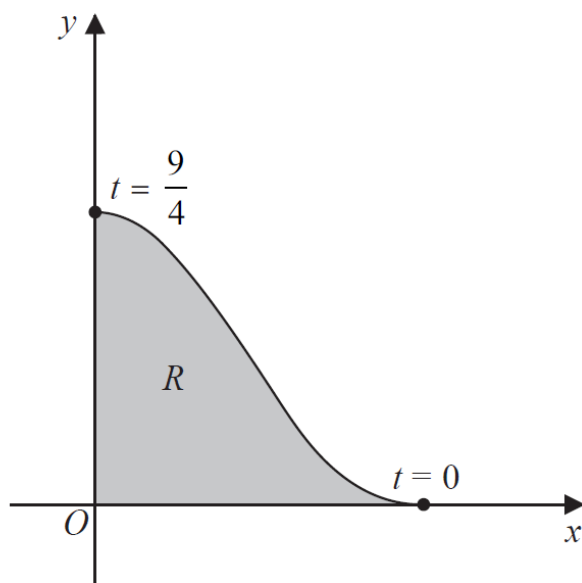


Figure 2

Figure 2 shows a sketch of the curve with parametric equations

$$x = \sqrt{9 - 4t} \quad y = \frac{t^3}{\sqrt{9 + 4t}} \quad 0 \leq t \leq \frac{9}{4}$$

The curve touches the x -axis when $t = 0$ and meets the y -axis when $t = \frac{9}{4}$

The region R , shown shaded in Figure 2, is bounded by the curve, the x -axis and the y -axis.

(a) Show that the area of R is given by

$$K \int_0^{\frac{9}{4}} \frac{t^3}{\sqrt{81 - 16t^2}} dt$$

where K is a constant to be found.

(4)

(b) Using the substitution $u = 81 - 16t^2$, or otherwise, find the exact area of R .

(Solutions relying on calculator technology are not acceptable.)

(6)

7.

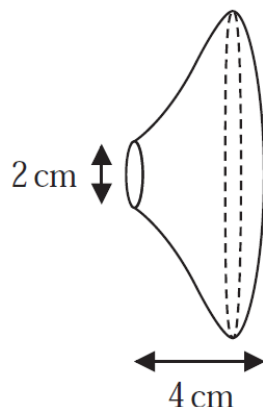


Figure 3

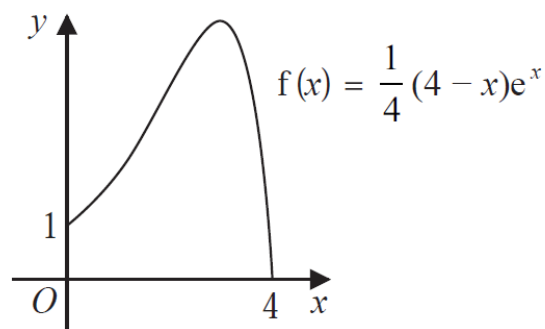


Figure 4

Figure 3 shows the design of a doorknob.

The shape of the doorknob is formed by rotating the curve shown in Figure 4 through 360° about the x -axis, where the units are centimetres.

The equation of the curve is given by

$$f(x) = \frac{1}{4}(4 - x)e^x \quad 0 \leq x \leq 4$$

- (a) Show that the volume, $V \text{ cm}^3$, of the doorknob is given by

$$V = K \int_0^4 (x^2 - 8x + 16)e^{2x} dx$$

where K is a constant to be found.

(3)

- (b) Hence, find the exact value of the volume of the doorknob.

Give your answer in the form $p\pi(e^q + r) \text{ cm}^3$ where p , q and r are simplified rational numbers to be found.

(5)

2022 January P4 Q9

9. (a) Find the derivative with respect to y of

$$\frac{1}{(1 + 2\ln y)^2}$$

(2)

- (b) Hence find a general solution to the differential equation

$$3\operatorname{cosec}(2x) \frac{dy}{dx} = y(1 + 2\ln y)^3 \quad y > 0 \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$$

(4)

- (c) Show that the particular solution of this differential equation for which $y = 1$ at $x = \frac{\pi}{6}$ is given by

$$y = e^{A\sec x - \frac{1}{2}}$$

where A is an irrational number to be found.

(5)

2022 June P4 Q2

2. (a) Express $\frac{1}{(1+3x)(1-x)}$ in partial fractions. (3)

(b) Hence find the solution of the differential equation

$$(1+3x)(1-x)\frac{dy}{dx} = \tan y \quad -\frac{1}{3} < x \leq \frac{1}{2}$$

for which $x = \frac{1}{2}$ when $y = \frac{\pi}{2}$

Give your answer in the form $\sin^n y = f(x)$ where n is an integer to be found.

(6)

2022 June P4 Q5

5. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Use the substitution $x = 2 \sin u$ to show that

$$\int_0^1 \frac{3x + 2}{(4 - x^2)^{\frac{3}{2}}} dx = \int_0^p \left(\frac{3}{2} \sec u \tan u + \frac{1}{2} \sec^2 u \right) du$$

where p is a constant to be found.

(4)

(b) Hence find the exact value of

$$\int_0^1 \frac{3x + 2}{(4 - x^2)^{\frac{3}{2}}} dx$$

(4)

2022 June P4 Q8

8.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

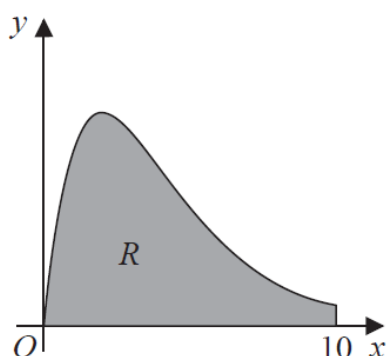


Figure 2

Figure 2 shows the curve with equation

$$y = 10xe^{-\frac{1}{2}x} \quad 0 \leq x \leq 10$$

The finite region R , shown shaded in Figure 2, is bounded by the curve, the x -axis and the line with equation $x = 10$

The region R is rotated through 2π radians about the x -axis to form a solid of revolution.

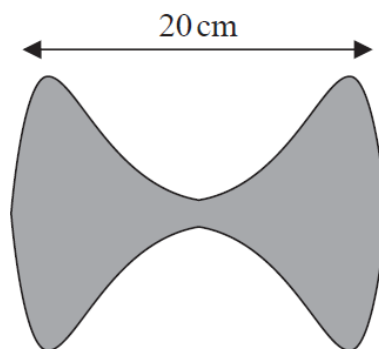


Figure 3

(a) Show that the volume, V , of this solid is given by

$$V = k \int_0^{10} x^2 e^{-x} dx$$

where k is a constant to be found.

(2)

(b) Find $\int x^2 e^{-x} dx$

(3)

Figure 3 represents an exercise weight formed by joining two of these solids together.

The exercise weight has mass 5 kg and is 20 cm long.

Given that

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

and using your answers to part (a) and part (b),

(c) find the density of this exercise weight. Give your answer in grams per cm^3 to 3 significant figures.

(5)

2022 October P4 Q2

2. (a) Express $\frac{3x}{(2x-1)(x-2)}$ in partial fraction form.

(3)

- (b) Hence show that

$$\int_5^{25} \frac{3x}{(2x-1)(x-2)} dx = \ln k$$

where k is a fully simplified fraction to be found.

(Solutions relying entirely on calculator technology are not acceptable.)

(4)

2022 October P4 Q5

5.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

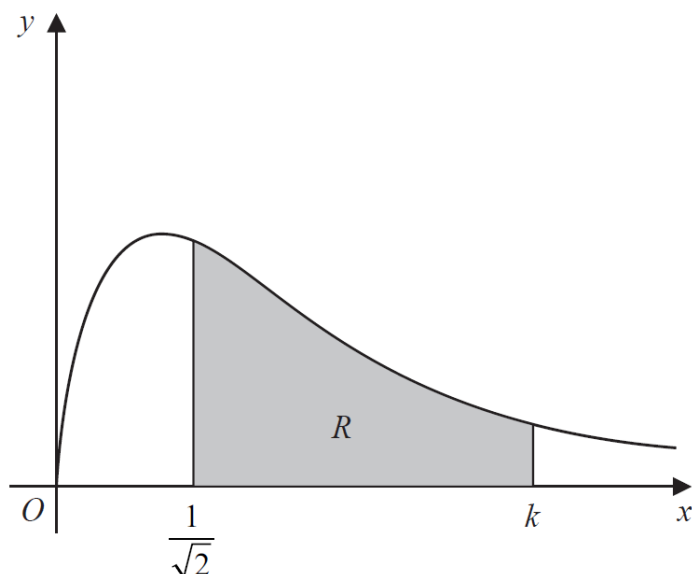


Figure 2

Figure 2 shows a sketch of part of the curve with equation

$$y = \frac{12\sqrt{x}}{(2x^2 + 3)^{1.5}}$$

The region R , shown shaded in Figure 2, is bounded by the curve, the line with equation $x = \frac{1}{\sqrt{2}}$, the x -axis and the line with equation $x = k$.

This region is rotated through 360° about the x -axis to form a solid of revolution.

Given that the volume of this solid is $\frac{713}{648}\pi$, use algebraic integration to find the exact value of the constant k .

(6)

2022 October P4 Q7

7.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

- (i) Use the substitution $u = e^x - 3$ to show that

$$\int_{\ln 5}^{\ln 7} \frac{4e^{3x}}{e^x - 3} dx = a + b \ln 2$$

where a and b are constants to be found.

(7)

- (ii) Show, by integration, that

$$\int 3e^x \cos 2x \, dx = pe^x \sin 2x + qe^x \cos 2x + c$$

where p and q are constants to be found and c is an arbitrary constant.

(5)

2023 January P4 Q3

3.

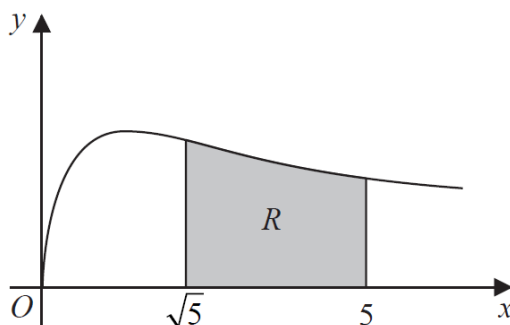


Figure 1

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Figure 1 shows a sketch of the curve with equation

$$y = \sqrt{\frac{3x}{3x^2 + 5}} \quad x \geq 0$$

The finite region R , shown shaded in Figure 1, is bounded by the curve, the x -axis and the lines with equations $x = \sqrt{5}$ and $x = 5$

The region R is rotated through 360° about the x -axis.

Use integration to find the exact volume of the solid generated. Give your answer in the form $a \ln b$, where a is an irrational number and b is a prime number.

(5)

2023 January P4 Q4

4. (a) Using the substitution $u = \sqrt{2x+1}$, show that

$$\int_4^{12} \sqrt{8x+4} e^{\sqrt{2x+1}} dx$$

may be expressed in the form

$$\int_a^b k u^2 e^u du$$

where a , b and k are constants to be found.

(4)

- (b) Hence find, by algebraic integration, the exact value of

$$\int_4^{12} \sqrt{8x+4} e^{\sqrt{2x+1}} dx$$

giving your answer in simplest form.

(5)

2023 June P4 Q3

3.
$$f(x) = \frac{8x - 5}{(2x - 1)(4x - 3)} \quad x > 1$$

(a) Express $f(x)$ in partial fractions.

(3)

(b) Hence find $\int f(x) dx$

(3)

(c) Use the answer to part (b) to find the value of k for which

$$\int_k^{3k} f(x) dx = \frac{1}{2} \ln 20$$

(5)

2023 June P4 Q5

5. (i) Find

$$\int x^2 e^x dx$$

(4)

(ii) Use the substitution $u = \sqrt{1 - 3x}$ to show that

$$\int \frac{27x}{\sqrt{1 - 3x}} dx = -2(1 - 3x)^{\frac{1}{2}}(Ax + B) + k$$

where A and B are integers to be found and k is an arbitrary constant.

(6)

2023 June P4 Q6

6.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The temperature, $\theta^{\circ}\text{C}$, of a car engine, t minutes after the engine is turned off, is modelled by the differential equation

$$\frac{d\theta}{dt} = -k(\theta - 15)^2$$

where k is a constant.

Given that the temperature of the car engine

- is 85°C at the instant the engine is turned off
- is 40°C exactly 10 minutes after the engine is turned off

(a) solve the differential equation to show that, according to the model

$$\theta = \frac{at + b}{ct + d}$$

where a , b , c and d are integers to be found.

(7)

(b) Hence find, according to the model, the time taken for the temperature of the car engine to reach 20°C . Give your answer to the nearest minute.

(2)

2023 October P4 Q3

3. In this question you must show all stages of your working.

Solutions based on calculator technology are not acceptable.

- (i) Use integration by parts to find the exact value of

$$\int_0^4 x^2 e^{2x} \, dx$$

giving your answer in simplest form.

(5)

- (ii) Use integration by substitution to show that

$$\int_3^{\frac{21}{2}} \frac{4x}{(2x-1)^2} \, dx = a + \ln b$$

where a and b are constants to be found.

(7)

2023 October P4 Q7

7. The number of goats on an island is being monitored.

When monitoring began there were 3000 goats on the island.

In a simple model, the number of goats, x , in thousands, is modelled by the equation

$$x = \frac{k(9t + 5)}{4t + 3}$$

where k is a constant and t is the number of years after monitoring began.

(a) Show that $k = 1.8$

(2)

(b) Hence calculate the long-term population of goats predicted by this model.

(1)

In a **second** model, the number of goats, x , in thousands, is modelled by the differential equation

$$3 \frac{dx}{dt} = x(9 - 2x)$$

(c) Write $\frac{3}{x(9 - 2x)}$ in partial fraction form.

(3)

(d) Solve the differential equation with the initial condition to show that

$$x = \frac{9}{2 + e^{-3t}}$$

(5)

(e) Find the long-term population of goats predicted by this **second** model.

(1)

2024 January P4 Q2

2. Given that

$$\frac{3x + 4}{(x - 2)(2x + 1)^2} \equiv \frac{A}{x - 2} + \frac{B}{2x + 1} + \frac{C}{(2x + 1)^2}$$

(a) find the values of the constants A , B and C .

(4)

(b) Hence find the exact value of

$$\int_7^{12} \frac{3x + 4}{(x - 2)(2x + 1)^2} dx$$

giving your answer in the form $p \ln q + r$ where p , q and r are rational numbers.

(6)

2024 January P4 Q5

5. (a) Find $\int x^2 \cos 2x \, dx$

(4)

(b) Hence solve the differential equation

$$\frac{dy}{dt} = \left(\frac{t \cos t}{y} \right)^2$$

giving your answer in the form $y^n = f(t)$ where n is an integer.

(5)

2024 January P4 Q7

7. (a) Using the substitution $u = 4x + 2 \sin 2x$, or otherwise, show that

$$\int_0^{\frac{\pi}{2}} e^{4x+2\sin 2x} \cos^2 x \, dx = \frac{1}{8}(e^{2\pi} - 1)$$

(5)

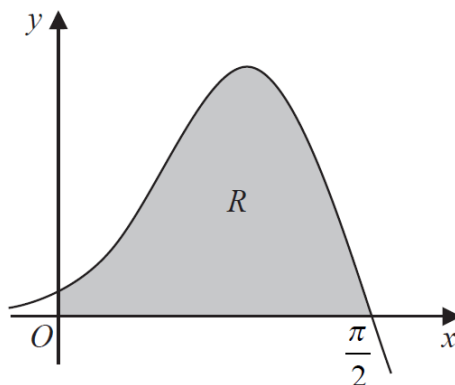


Figure 3

The curve shown in Figure 3, has equation

$$y = 6e^{2x+\sin 2x} \cos x$$

The region R , shown shaded in Figure 3, is bounded by the positive x -axis, the positive y -axis and the curve.

The region R is rotated through 2π radians about the x -axis to form a solid.

- (b) Use the answer to part (a) to find the volume of the solid formed, giving the answer in simplest form.

(3)